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Evaluation of artificial intelligence-based cephalometric tracing versus semi-automatic and manual tracing

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Abstract

Background Artificial intelligence (AI)-based cephalometric tracing has emerged as a promising tool that reduces operator variability and offers standardized, rapid, and reproducible assessments. This study aimed to evaluate the reliability and accuracy of three cephalometric tracing methods: AI-based automatic digital tracing, semi-automatic digital tracing, and manual digital tracing.

Materials & methods This study analyzed pre-treatment lateral cephalograms from 120 patients, comparing AI-based automatic digital tracing, semi-automatic digital tracing against manual digital tracing, which served as the control group. Angular and linear measurements were used as primary parameters, with 34 cephalometric measurements derived from 20 skeletal (SK) and 10 soft tissue (ST) landmarks. For parametric data, comparisons among the three methods were conducted using a repeated measures ANOVA test, while non-parametric data were analyzed using the Friedman test. A significance level of $P \leq 0.05$ was applied to all statistical analyses.

Results In maxillary skeletal measurements, SNA (°) showed significant variation, with the Automatic method reporting slightly higher mean values ($P < 0.001$). Similarly, mandibular skeletal measurements revealed a small but significant difference in SNB (°) values ($P = 0.002$), while MP-SN (°) showed no notable differences ($P = 0.118$). For inter-maxillary measurements, ANB (°) displayed significant differences ($P < 0.001$), with the Automatic method reporting higher mean values. Vertical measurements, such as GO-GN to SN (°) and the Gonial Angle (°), also demonstrated significant variations ($P < 0.001$), with manual methods generally reporting higher values. In dental measurements, L1 to NB (°) differed significantly across methods ($P = 0.022$), and U1 to NA measurements showed both angular ($P = 0.006$) and linear ($P < 0.001$) significant differences, with Automatic methods tending to report lower medians. Lastly, soft tissue measurements, particularly the nasolabial angle (°), exhibited significant differences ($P < 0.001$), with the Manual method recording the highest mean values.

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Conclusions AI-based automatic tracing tended to overestimate certain skeletal values, while manual methods showed greater consistency. Dental measurements were largely comparable across methods. The semi-automatic approach provided a practical balance between accuracy and efficiency, indicating potential for clinical application with further refinement.

Keywords Cephalometric analysis, Computerized tracing methods, Artificial intelligence, Digital cephalometry, Craniofacial measurements, Skeletal landmarks, Automated cephalometric software

Introduction

Cephalometry involves measuring and analyzing skull dimensions using standardized radiographs of the living head. Since its introduction by Broadbent and Hofrath in the 1930 s, it has evolved into an essential tool in orthodontics and craniofacial analysis for identifying specific landmarks on the skull, which enables the calculation of angular and linear measurements [1]. Cephalometric landmarks are specific points that indicate distinct features of the skull's soft and hard tissues. They are used to measure angles in degrees and distances in millimeters for cephalometric analysis [2]. Repeated cephalometric radiographs enable the analysis and prediction of growth, aid in treatment evaluation, and monitor changes between pre- and post-treatment stages. However, despite their widespread use in orthodontics, this method has significant limitations, including being time-consuming, prone to errors, and susceptible to systemic inaccuracies [3].

Artificial Intelligence (AI) technology allows computer programs and smartphone apps to automatically recognize landmarks and take measurements once a digital radiograph is uploaded (e.g., WebCeph web program) [4–7]. Computer-assisted cephalometric analysis automatically calculates angles and distances, minimizing errors that can arise from manually drawing lines between landmarks and measuring with a protractor [8, 9]. This advancement enables a more precise representation of anatomical structures for analysis and assists in the superimposition of images [10].

AI-based tracing programs revealed inconsistencies when compared to measurements from human experts and demonstrated a lack of reliability in assessing crucial cephalometric parameters. The automated WebCeph faces several challenges, including inadequate landmark identification and significant measurement inaccuracies [5, 11]. Manual oversight from the observer is essential to minimize the margin of error in fully automated cephalometric software utilizing artificial intelligence [12].

Conversely, some studies suggest that the results from fully automated cephalometric analysis software using artificial intelligence algorithms are similar to those of non-automated cephalometric analysis software [4, 13]. There is no consensus on the precision of these methods. Consequently, this cross-sectional analytical study aimed to evaluate the reliability and accuracy of three tracing techniques: automatic, semi-automatic, and manual digital.

Materials and methods

The study utilized a retrospective and consecutive sampling approach. A total of 120 pre-treatment lateral cephalograms were selected from patients who underwent orthodontic treatment at the Orthodontic Graduate Clinic, Faculty of Dentistry, Sana'a University, and were imaged at the Al Waleed Digital X-Ray Center between January 2022 and May 2023. Inclusion criteria included: availability of high-quality pre-treatment lateral cephalograms without artifacts, presence of a complete set of teeth up to the second molars, and clear visibility of all anatomical landmarks necessary for cephalometric analysis. Exclusion criteria included patients with impacted or unerupted teeth, missing incisors, craniofacial deformities, or excessive soft tissue that could obscure anatomical landmarks. Additionally, radiographs with any artifacts or poor image quality were excluded from the analysis. The same protocol was used for all 120 of the selected lateral cephalograms. The Ethical and Research Committees of Medical Research at Sana'a University approved this study. Furthermore, this research was designed according to STROBE regulations and conducted following the Helsinki Declaration. In addition, informed consent was obtained from all participating patients to clarify the purpose of the radiographs. For participants under the age of 18, consent was obtained from their parents or legal guardians to clarify the purpose of the radiographs.

Data processing and calibration

Before digitizing the images, an assessment of intra- and inter-examiner reliability was conducted. The cephalograms were randomly assigned to four orthodontic residents who received calibration training from an expert orthodontist to ensure measurement consistency. For inter-examiner reliability, each investigator (the residents and the expert orthodontist) measured ten radiographs using Oneceph. To evaluate intra-examiner reliability, each resident re-measured the same radiographs on two separate occasions with a four-week interval. To minimize errors related to fatigue, each resident completed a maximum of five daily tracings.

Measurement parameters

This study employed a combination of angular and linear measurements to evaluate craniofacial structures comprehensively. The parameters assessed included maxillary and mandibular skeletal measurements, intermaxillary relationships, vertical dimensions, dental positions, and soft tissue landmarks. These measurements provide valuable insight into skeletal alignment, dental positioning, soft tissue contours, and their interrelationships.

Skeletal parameters included

SNA (°) – Sella-Nasion-A Point angle.

A-N (mm)– Linear distance from A Point to Nasion.

SNB (°) – Sella-Nasion-B Point angle.

MP-SN (°) – Mandibular Plane to Sella-Nasion angle.

Pg to N (mm) – Linear distance from Pogonion to Nasion.

ANB (°) – A Point-Nasion-B Point angle.

Go-Gn-SN (°)– Gonion-Gnathion to Sella-Nasion angle.

Dental measurements included

L1 to NB (° and mm)– Angle and distance of the lower incisor relative to the Nasion-B Point line.

U1 to NA (° and mm) – Angle and distance of the upper incisor relative to the Nasion-A Point line.

Soft tissue analysis focused on the naso-labial angle (°)

All measurements were recorded in either degrees (°) for angular values or millimeters (mm) for linear values.

Tracing methods

Automatic digital tracing method

Digital images of each conventional lateral cephalogram were created by transferring the selected cephalogram files to the WebCeph software. Calibration was conducted by digitizing two points on the cephalostat's built-in ruler. Cephalometric landmarks were then identified and digitized automatically by the software, which generated the required lines and planes. Image enhancement features—such as magnification, brightness, and contrast adjustments—were utilized to improve landmark identification (Fig. 1A).

Semi-automatic digital tracing method

Digital images of each conventional lateral cephalogram were imported by transferring the selected files from the cephalogram archive folder to the Web Ceph software. All images were calibrated by digitizing two specific reference points on the built-in ruler within the Cephalostat. Cephalometric landmarks were then identified and digitized on-screen with semi-automated assistance, followed by manual corrections by orthodontic specialists to enhance accuracy. Orthodontic experts also evaluated the program's accuracy and performance during these procedures. The program automatically generated lines and planes based on the identified landmarks (Fig. 1B).

Manual digital tracing

A Samsung Galaxy Note 20 smartphone (Android) was utilized for cephalometric analysis through the OneCeph mobile application (version beta 7), created by NXS and designed by Dr. M. Pavan Kumar. Each

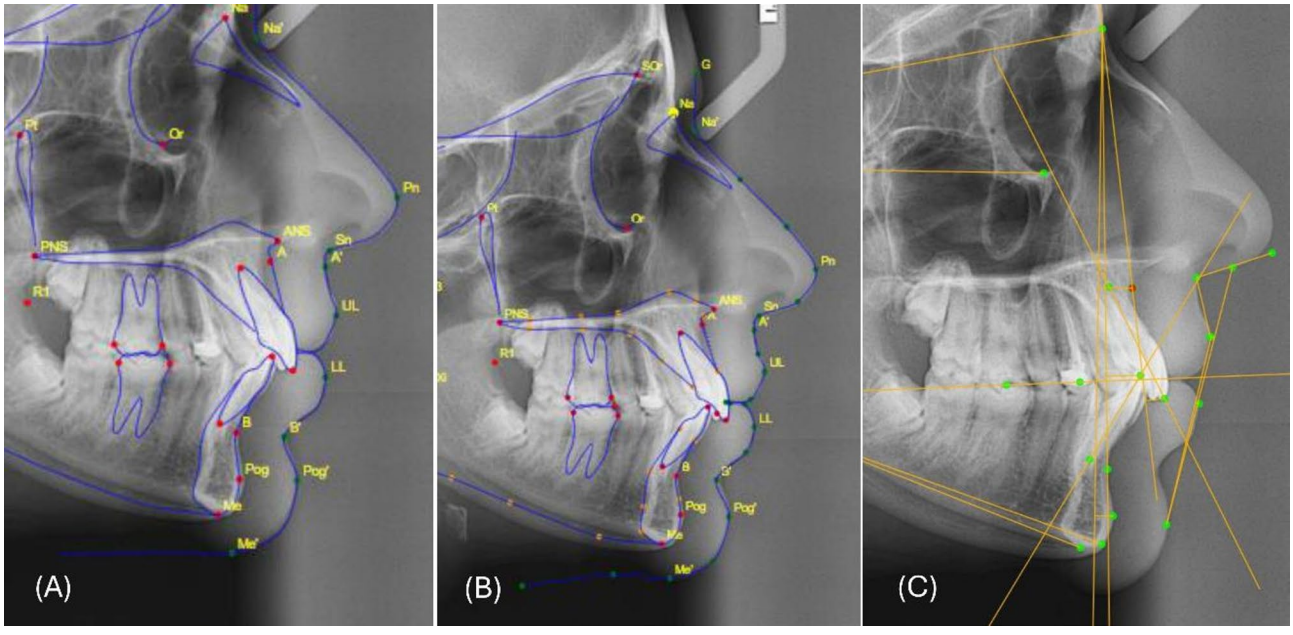

Fig. 1 Tracing methods, (**A**) Automatic Digital Tracing Method, (**B**) Semi-automatic Digital Tracing Method, and (**C**) Manual Digital Tracing

cephalometric image received a unique identification number for tracking purposes. Image calibration was performed by selecting a known reference distance on the built-in ruler visible on the cephalogram. This ensured that all measurements were scaled appropriately before landmark identification and analysis. Favorites were set in the toolbar, with options configured for various cephalometric analyses, including Steiner's, Downs', Holdaway's, Jarabak's, McNamara's, and Tweed's analyses. All selected cephalometric parameters were measured following these analyses, and the results were compiled for statistical analysis (Fig. 1C).

Statistical analysis

Statistical analysis was performed using IBM SPSS Statistics for Windows, Version 23.0 (IBM Corp., Armonk, NY, USA). Numerical data were evaluated for normality by examining the data distribution and conducting Kolmogorov-Smirnov and Shapiro-Wilk tests. Most data showed a normal (parametric) distribution, except for certain measurements—A to N (mm), Pg to N (mm), ANB (°), U1 to NA (mm), U1 to NA (°), and L1 to NB (mm)—which exhibited a non-normal (non-parametric) distribution. Data were summarized as mean, standard deviation (SD), median, and range values. For parametric data, comparisons across the three modalities were made using a repeated measures ANOVA test. Bonferroni's post-hoc test was applied for pairwise comparisons with significant ANOVA results. For non-parametric data, the Friedman test was utilized to compare the modalities, followed by Dunn's test for pairwise comparisons when the Friedman test indicated significance. A significant level of $P \leq 0.05$ was established for all analyses.

Results

The present study was conducted on 120 subjects: 71 females (59.2%) and 49 males (40.8%). The mean and standard deviation values for ages were 20.3 ± 3.9 years old, with a minimum age of 14 and a maximum age of 33.

Regarding the skeletal measurements across three methods (Table 1). For maxillary measurements, the automatic method yielded significantly higher SNA angles (82.09 ± 3.78°) compared to semi-automatic (80.88 ± 3.76°) and manual (80.8 ± 3.83°) methods ($P < 0.001$), suggesting a potential overestimation of maxillary protrusion. Similarly, the automatic method showed greater A-N distances (median: 1.25 mm) than semi-automatic (1.11 mm) and manual (1.1 mm) measurements ($P < 0.001$), though all methods exhibited wide variability.

In mandibular assessments, the automatic method produced slightly higher SNB angles (76.01 ± 3.42°) than semi-automatic (75.82 ± 3.54°) and manual (75.7 ± 3.51°) methods ($P = 0.002$), while MP-SN angles showed no significant differences ($P = 0.118$). The automatic method also recorded larger ANB angles (6.06 ± 2.34°) compared to semi-automatic (5.15 ± 2.57°) and manual (5.1 ± 2.5°) methods ($P < 0.001$).

Vertical measurements revealed further discrepancies: manual methods produced higher GO-GN-SN angles (34.97 ± 6.26°) and gonial angles (126.26 ± 6.78°) than automatic methods (33.44 ± 6.12° and 120.34 ± 7°, respectively; $P < 0.001$).

Regarding the dental measurements, the inter-incisal angle showed no significant differences among automatic (124.93 ± 10.57°), semi-automatic (124.82 ± 15.88°), and manual (125.95 ± 11.1°) methods ($p = 0.373$), indicating high agreement for this parameter. However, the automatic method produced a slightly higher L1 to NB angle (28.33 ± 6.84°) compared to semi-automatic (27.71 ± 7.09°) and manual (27.64 ± 7.36°) methods ($p = 0.022$). Similarly,

Table 1 Skeletal measurements across automatic, semi-automatic, and manual digital methods

Cephalometric parameters	Automatic method	Semi-automatic method	Manual digital method	*P*-value	Statistical test
Maxillary skeletal measurements:					
SNA (°) Mean ± SD	82.09 ± 3.78	80.88 ± 3.76	80.8 ± 3.83	< 0.001*	ANOVA test
A-N (mm) Median (Min – Max)	1.25 (−5.17–7.66)	1.11 (−5.87–8.33)	1.1 (−6–8.8)	< 0.001*	Friedman's test
Mandibular skeletal measurements:					
SNB (°) Mean ± SD	76.01 ± 3.42[A]	75.82 ± 3.54[B]	75.7 ± 3.51[B]	0.002*	ANOVA test
MP-SN (°) Mean ± SD	16.47 ± 5.16	16.03 ± 4.93	15.91 ± 4.85	0.118	ANOVA test
Pg to N(mm) Median (Min – Max)	−6.21 (−18.93–7.97)	−5.98 (− 21.34, 30.48)	−5.9 (−21.8–8.8)	0.991	Friedman's test
Intermaxillary measurement:					
ANB (°)Mean ± SD	6.06 ± 2.34	5.15 ± 2.57	5.1 ± 2.5	< 0.001*	Friedman's test
Vertical measurements					
GO - GN – SN (°)Mean ± SD	33.44 ± 6.12	34.33 ± 6.21	34.97 ± 6.26	< 0.001*	ANOVA test
Gonial angle (°)Mean ± SD	120.34 ± 7	125.18 ± 6.92	126.26 ± 6.78[A]	< 0.001*	ANOVA test

Abbreviations: *SNA* Sella-Nasion-A point angle, *A-N* A point to Nasion, *SNB* Sella-Nasion-B point angle, *MP-SN* Mandibular Plane to Sella-Nasion angle, *Pg to N* Pogonion to Nasion, *ANB* A point-Nasion-B point angle, *Go-Gn-SN* Gonion-Gnathion to Sella-Nasion angle), *(°)* Degree, *(mm)* Millimeters

*highly significant

Table 2 Dental and soft tissue measurements across automatic, semi-automatic, and manual digital methods

Cephalometric parameters	Automatic method	Semi-automatic method	Manual Method	*P*-value	Statistical test
Dental measurements					
Inter-incisal (°) Mean ± SD	124.93± 10.57	124.82±15.88	125.95±11.1	0.373	ANOVA test
L1 to NB (°) Mean ± SD	28.33 ± 6.8[A]	27.71±7.09[B]	27.64±7.36[B]	0.022	ANOVA test
U1 to NA (°) Mean ± SD	20.8±6.94	21.05±7.46	21.21±7.43	0.006	Friedman's test
U1 to NA (mm) Median (Min – Max)	2.62 (0.09 –11.89)	4.08 (–2.05-11.35)	4.15 (–5.2-12.5)	<0.001	Friedman's test
L1 to NB (mm) Median (Min – Max)	6.01 (0.2-14.93)	5.86 (0.6-26.98)	6 (–0.7- 13.7)	0.053	Friedman's test
Soft tissue measurements:					
Naso-labial angle (°) Mean ± SD	103.64±12.84	106.37±12.01	107.78±11.77	<0.001	ANOVA test

Abbreviations: *L1 to NB* Lower Incisor to Nasion-B point line, *U1 to NA* Upper Incisor to Nasion-A point) line, *(°)* Degree, *(mm)* Millimeters

U1 to NA angles differed significantly ($p = 0.006$), though values were clinically close (automatic: 20.8 ± 6.94°; manual: 21.21 ± 7.43°). Notably, U1 to NA linear measurements (mm) varied markedly by method ($p < 0.001$), with the automatic method yielding a lower median (2.62 mm) than semi-automatic (4.08 mm) and manual (4.15 mm) methods (Table 2).

Regarding the soft tissue (Table 2), the naso-labial angle also differed significantly ($p < 0.001$), with manual measurements (107.78 ± 11.77°) exceeding automatic values (103.64 ± 12.84°).

Discussion

The process of point identification and measurement generation in cephalometric analysis remains labor-intensive and time-consuming. In response, recent technological advancements, particularly those involving AI, have enabled the development of novel tools designed to expedite and streamline this process [14]. Three commercially available AI-based lateral cephalometric tracing software programs were assessed: WebCeph™, Cephio, and Cepro DDH Inc., alongside a semi-automatic program from Dolphin Imaging Systems LLC. All three AI-based tracing software programs exhibited inaccuracies compared to measurements from human experts and failed to demonstrate reliability in assessing key cephalometric parameters [15]. A significant statistical difference (> 2 units) was observed in the mean measurements between the human expert and WebCeph™ across all parameters. The automated WebCeph has several problems, including poor landmark identification and substantial measurement errors [5, 11]. Therefore, the current study aimed to assess the reliability and accuracy of three tracing methods: automatic, semi-automatic, and manual cephalometric tracing.

In the analysis of skeletal maxillary dimensions, the automatic method produced significantly higher SNA angles than both semi-automatic and manual methods ($P < 0.001$), indicating a tendency to overestimate maxillary protrusion. Similarly, A-N linear distances were greater in the automatic group ($P < 0.001$), although all methods displayed broad variability. These outcomes suggest a potential misclassification risk of sagittal skeletal patterns, possibly stemming from inaccurate localization of point A—a well-documented limitation in both manual and AI-assisted tracing methods.

For mandibular skeletal assessments, the automatic method yielded slightly elevated SNB angles compared to other methods ($P = 0.002$), while MP-SN angles did not significantly differ ($P = 0.118$). The automatic method also reported higher ANB angles ($P < 0.001$), which may exaggerate skeletal discrepancies. Manual tracings, in contrast, produced higher GO-GN-SN and gonial angles than the automatic approach ($P < 0.001$), suggesting greater accuracy in evaluating vertical skeletal relationships through manual identification.

A study found that fully automated AI cephalometric analysis closely matches those from non-automated cephalometric analysis software despite variations in specific parameters. Observer manual intervention is essential to reduce errors in AI-based fully automatic cephalometric software [12]. Landmarks were identified manually for the semi-automatic cephalometric analysis program, which then calculated distances and angles automatically. While this process minimizes inaccuracies associated with entirely manual measurements, errors in landmark identification can still arise, as the operator's experience influences random errors [16]. Additionally, challenges in locating point A have been reported before and could clarify the tracing variance [17, 18]. Interestingly, AI-driven cephalometric programs have been

noted for incorrectly identifying point A, which results in the misclassification of sagittal skeletal patterns [19]. This is not unexpected, since AI programs' performance is validated using human-labelled training data.

In dental measurements, consistency was observed for the inter-incisal angle, with no significant differences across all tracing methods ($P = 0.373$). However, the automatic method demonstrated slightly elevated L1 to NB angles ($P = 0.022$) and differing U1 to NA angles ($P = 0.006$), though the clinical implications remain minimal. Marked differences in U1 to NA linear distances ($P < 0.001$) and nasolabial angle values ($P < 0.001$) highlight the potential inaccuracies of AI methods in tracing soft tissue landmarks. For example, WebCeph™ demonstrated a notably higher measurement error (mean deviation of 9.3) for the nasolabial angle compared to previously reported values, likely due to the complexity of soft tissue curvature and difficulties in landmark placement along large-radius curves.

A study found that the error magnitude of WebCeph™ measurements for the nasolabial angle was greater at 9.3, compared to the previously reported value of 2.7 ± 4.9 [20]. These errors might stem from the nasolabial angle landmarks on a larger radius curve [21]. A study found that the automatic WebCeph presents various issues concerning soft tissue tracing [5]. Significant variations were noted in the angular measurements of the L1-MP angle values produced by WebCeph™, likely due to irregularities in the mandibular plane identification [15].

Various studies have indicated that certain cephalometric landmarks, notably the Menton, Gonion, lower incisor apex, Orbitale, Porion, and Nasion, are unreliable and prone to inaccuracies, regardless of whether a manual, semi-automatic, or fully automatic method is used for their measurement identification [18, 22, 23].

In contrast, certain studies indicated that the outcomes from fully automated cephalometric analysis software utilizing artificial intelligence algorithms are comparable to those from non-automated cephalometric analysis software [4, 6, 13].

In summary, while there have been advancements, fully automated AI-based cephalometric tracing algorithms have not yet reached the reliability needed to supplant the traditional manual method. AI technologies often face challenges with consistent landmark identification on lateral cephalometric radiographs. However, WebCeph shows improved performance when paired with manual adjustments (semi-automatic WebCeph), providing results that are on par with traditional software but with reduced processing times.

Nonetheless, AI analyses should still be approached with caution and always require final oversight from a skilled orthodontist. This is consistent with the results of many previous studies [5, 14, 15, 24–26].

The current study presents some limitations. Firstly, relying on cephalograms from only one center could restrict the generalizability of the results, owing to possible regional or demographic biases. Secondly, the results pertain specifically to the software utilized (WebCeph for automated tracing and OneCeph for manual/semi-automated tracing) and might not extend to other AI-driven cephalometric tools. Lastly, the soft tissue analysis focused solely on the nasolabial angle, leaving out other essential soft tissue parameters.

Conclusion

The automatic method generally produced higher maxillary and mandibular skeletal measurements compared to semi-automatic and manual methods, suggesting a tendency to overestimate certain parameters, particularly SNA, ANB angles, and A-N distances. Vertical skeletal measurements and soft tissue assessments showed notable discrepancies, with manual methods yielding higher values. Dental measurements demonstrated closer agreement across methods, with minor but statistically significant differences in some parameters. Moreover, the semi-automatic approach offers a reliable compromise, balancing automation with manual precision. With further refinement, automated techniques hold strong potential as dependable clinical tools.

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Authors' contributions
S.K.Z. and O.A.I.A. Conceptualization, Data curation, Methodology, Statistical analysis, Writing- Original draft preparation, Review, and editing, Final draft approval. S.A.E.: Responsible for reviewing, editing, final draft approval prior to submission, and responding to reviewers' comments.All other authors approved the final paper after undergoing critical revisions.

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Data availability
The datasets used and analyzed during the study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
The Ethical and Research Committees of the Medical Research at Sana'a University approved this study. Furthermore, this research was designed following STROBE regulations and carried out following the Helsinki Declaration. In addition, informed consent was obtained from all participating patients. For participants under the age of 18, consent was obtained from their parents or legal guardians to clarify the purpose of the radiographs.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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